



## Supplemental S1 — Literature Review (auto-composed from deep search)

## Supplemental S1 — Literature Review (auto-composed from deep search) I

*Composed by `scripts/s_compose_literature_review.py` from the most recent deep-search run. Edit the script, not this file — manual edits will be overwritten on the next pipeline run.*

This review covers **3 keyword(s)**, **300 unique paper(s)** (retrieved at up to 100 per keyword from arxiv, crossref). All references are stored in `manuscript/references_deep.bib` and resolved by the combined-PDF pipeline (`Pandoc --natbib + BibTeX` over all `manuscript/*.bib` files).

# convex optimization I

*Papers retrieved: 100 · per-source contributions: arxiv=100,  
crossref=100 · backend errors: none*

| Cite                     | Title                                                   | Year DOI / URL                                    |
|--------------------------|---------------------------------------------------------|---------------------------------------------------|
| [@anon2004convex]        | Convex optimization problems                            | 2004 DOI 10.1017/cbo9780511804441.005 (open)      |
| [@anon2004convexa]       | Convex functions                                        | 2004 DOI 10.1017/cbo9780511804441.004 (open)      |
| [@anonreverse]           | Reverse Convex Optimization, Reverse convex programming | n.d. DOI 10.1007/springerreference_72642 (open)   |
| [@suh20221]              | 1. Convex Optimization Basics                           | 2022 DOI 10.1561/9781638280538.ch1 (open)         |
| [@anon2021convex]        | Convex Optimization and Efficiency                      | 2021 DOI 10.1017/9781108699211.006 (open)         |
| [@anon2021ellipsoid]     | Ellipsoid Method for Convex Optimization                | 2021 DOI 10.1017/9781108699211.015 (open)         |
| [@nesterov2004smooth]    | Smooth Convex Optimization                              | 2004 DOI 10.1007/978-1-4419-8853-9_2 (open)       |
| [@nesterov2004nonsmooth] | Nonsmooth Convex Optimization                           | 2004 DOI 10.1007/978-1-4419-8853-9_3 (open)       |
| [@anon2022convex]        | Convex Optimization Basics                              | 2022 DOI 10.1561/978-1-63828-053-820251002 (open) |
| [@anon2004convexb]       | Convex sets                                             | 2004 DOI 10.1017/cbo9780511804441.003 (open)      |

# convex optimization II

|                          |                                                                                                                                                    |                                                  |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| [@anon2011what]          | What Is Convex Optimization?                                                                                                                       | 2011 DOI 10.1201/b11156-7 (open)                 |
| [@anon2011convex]        | Convex Semi-Infinite Optimization                                                                                                                  | 2011 DOI 10.1201/b11156-17 (open)                |
| [@anon2011tools]         | Tools for Convex Optimization                                                                                                                      | 2011 DOI 10.1201/b11156-8 (open)                 |
| [@anon2004convexc]       | Convex Sets. Convex and Generalized Convex Functions                                                                                               | 2004 DOI 10.1016/b978-044450550-7/50002-8 (open) |
| [@anon20146]             | 6 Convex optimization algorithms                                                                                                                   | 2014 DOI 10.1515/9783110361629.117 (open)        |
| [@jacobsen2008reverse]   | Reverse Convex Optimization                                                                                                                        | 2008 DOI 10.1007/978-0-387-74759-0_564 (open)    |
| [@nesterov2018nonsmooth] | Nonsmooth Convex Optimization                                                                                                                      | 2018 DOI 10.1007/978-3-319-91578-4_3 (open)      |
| [@jacobsen2001bb]        | BB algorithm; Concave programming; D.C. programming; Quadratic knapsack. Quadratic programming with bound constraints; Reverse convex optimization | 2001 DOI 10.1007/0-306-48332-7_431 (open)        |
|                          | Standard quadratic optimization problems: Theory REVERSE CONVEX OPTIMIZATION                                                                       |                                                  |
| [@tuypartly]             | Partly Convex and Convex-Monotonic Optimization Problems                                                                                           | n.d. DOI 10.1007/3-540-27170-8_37 (open)         |
| [@nesterov2018smooth]    | Smooth Convex Optimization                                                                                                                         | 2018 DOI 10.1007/978-3-319-91578-4_2 (open)      |
| [@anon2011weak]          | Weak Sharp Minima in Convex Optimization                                                                                                           | 2011 DOI 10.1201/b11156-15 (open)                |

# convex optimization III

|                            |                                                                                          |                                                 |
|----------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------|
| [@anonconvex]              | Convex Discrete Optimization                                                             | n.d. DOI 10.1007/springerreference_72172 (open) |
| [@anon2025convex]          | Convex Functions                                                                         | 2025 DOI 10.1017/9781009510561.012 (open)       |
| [@zaslavski2020nonsmooth]  | Nonsmooth Convex Optimization                                                            | 2020 DOI 10.1007/978-3-030-60300-7_2 (open)     |
| [@anon2020appendix]        | Appendix:                                                                                | 2020 DOI 10.2307/j.ctvqsdxd.15 (open)           |
| [@anon2005connectedness]   | Connectedness of efficient points in convex and convex transformable vector optimization | 2005 DOI 10.1080/02331930500096270 (open)       |
| [@anon2025convexa]         | Convex Programming Problems and Convex Theorem of the Alternative                        | 2025 DOI 10.1017/9781009510561.021 (open)       |
| [@tuy1998convex]           | Convex Functions                                                                         | 1998 DOI 10.1007/978-1-4757-2809-5_2 (open)     |
| [@mattingley2009automatic] | Automatic code generation for real-time convex optimization                              | 2009 DOI 10.1017/cbo9780511804458.002 (open)    |
| [@anon2011toolsa]          | Tools for Convex Optimization                                                            | 2011 DOI 10.1201/b11156-3 (open)                |
| [@tuy1998convexa]          | Convex Sets                                                                              | 1998 DOI 10.1007/978-1-4757-2809-5_1 (open)     |
| [@anon2011whata]           | What Is Convex Optimization?                                                             | 2011 DOI 10.1201/b11156-2 (open)                |
| [@bonnans2019convex]       | A Convex Optimization Toolbox                                                            | 2019 DOI 10.1007/978-3-030-14977-2_1 (open)     |
| [@anon2020appendixa]       | Appendix: Executive Summary on Efficient Solvability of Convex Optimization Problems     | 2020 DOI 10.1515/9780691200316-013 (open)       |

# convex optimization IV

|                          |                                                      |
|--------------------------|------------------------------------------------------|
| [@anon20142]             | 2 Convex sets and convex functions                   |
| [@anon2021convexa]       | Convex Optimization                                  |
| [@anon2017convex]        | Convex Optimization Problems                         |
| [@zaslavski2020pdabased] | PDA-Based Method for Convex Optimization             |
| [@tuy2016convex]         | Convex Functions                                     |
| [@anon2025optimality]    | Optimality Conditions in Convex Programming          |
| [@anon2021convexb]       | Convex Functions                                     |
| [@li2015convex]          | Convex Relaxation                                    |
| [@lasserre2011convex]    | On convex optimization without convex representation |
| [@anon2021convexc]       | Convex Analysis and Convex Optimization              |
| [@anonconvexa]           | Convex Envelopes in Optimization Problems            |
| [@anon2004preface]       | Preface                                              |
| [@anon2025first]         | First Acquaintance with Convex Sets                  |

|          |                                        |
|----------|----------------------------------------|
| 2014 DOI | 10.1515/9783110361629.30 (open)        |
| 2021 DOI | 10.1017/9781108980647.012 (open)       |
| 2017 DOI | 10.1201/9781315366920-5 (open)         |
| 2020 DOI | 10.1007/978-3-030-37822-6_9 (open)     |
| 2016 DOI | 10.1007/978-3-319-31484-6_2 (open)     |
| 2025 DOI | 10.1017/9781009510561.024 (open)       |
| 2021 DOI | 10.1002/9781119804093.ch3 (open)       |
| 2015 DOI | 10.1007/978-3-662-46356-7_6 (open)     |
| 2011 DOI | 10.1007/s11590-011-0323-1 (open)       |
| 2021 DOI | 10.1017/9781108377447.033 (open)       |
| n.d. DOI | 10.1007/springerreference_72173 (open) |
| 2004 DOI | 10.1017/cbo9780511804441.001 (open)    |
| 2025 DOI | 10.1017/9781009510561.003 (open)       |

# convex optimization V

|                               |                                                                                                                                                         |                                                      |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| [@anon2021convexd]            | Convex Sets                                                                                                                                             | 2021 DOI<br>10.1002/9781119804093.ch2<br>(open)      |
| [@tuy2016convexa]             | Convex Sets                                                                                                                                             | 2016 DOI 10.1007/978-3-319-31484-6_1 (open)          |
| [@anon2004introduction]       | Introduction                                                                                                                                            | 2004 DOI<br>10.1017/cbo9780511804441.002<br>(open)   |
| [@singer1999duality]          | Duality in quasi-convex<br>supremization and reverse<br>convex infimization via<br>abstract convex<br>analysis, and applications<br>to approximation ** | 1999 DOI<br>10.1080/02331939908844436<br>(open)      |
| [@anon2025firsta]             | First Acquaintance with<br>Convex Functions                                                                                                             | 2025 DOI<br>10.1017/9781009510561.013<br>(open)      |
| [@singer2001duality]          | On Duality for<br>Quasi-convex<br>Supremization and<br>Reverse Convex<br>Infimization                                                                   | 2001 DOI 10.1007/978-1-4613-0295-7_16 (open)         |
| [@anon2004references]         | References                                                                                                                                              | 2004 DOI<br>10.1017/cbo9780511804441.016<br>(open)   |
| [@anonglobal]                 | Global Optimization:<br>Tight Convex<br>Underestimators                                                                                                 | n.d. DOI 10.1007/springerref-<br>erence_72325 (open) |
| [@marechal2001generatin<br>g] | Generating Convex<br>Functions                                                                                                                          | 2001 DOI 10.1007/978-1-4613-0279-7_21 (open)         |
| [@anon2025convexb]            | Convex Programming,<br>Lagrange Duality, Saddle<br>Points                                                                                               | 2025 DOI<br>10.1017/9781009510561.020<br>(open)      |
| [@peypouquet2015convex]       | Convex Analysis and<br>Subdifferential Calculus                                                                                                         | 2015 DOI 10.1007/978-3-319-13710-0_3 (open)          |



# convex optimization VI

|                             |                                                                                      |                                                              |
|-----------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------|
| [@anon2025convexc]          | Convex Programming in Cone-Constrained Form                                          | 2025 DOI<br>10.1017/9781009510561.023<br>(open)              |
| [@anon2025minima]           | Minima and Maxima of Convex Functions                                                | 2025 DOI<br>10.1017/9781009510561.015<br>(open)              |
| [@halman2015approximating]  | Approximating convex functions via non-convex oracles under the relative noise model | 2015 DOI<br>10.1016/j.disopt.2014.12.001<br>(open)           |
| [@stillwell2019statistical] | Statistical Inference via Convex Optimization                                        | 2019 DOI 10.23943/princeton/9780691197296.001.0001<br>(open) |
| [@dutta2014barrier]         | Barrier method in nonsmooth convex optimization without convex representation        | 2014 DOI 10.1007/s11590-014-0811-1 (open)                    |
| [@youness1999econvex]       | E-Convex Sets, E-Convex Functions, and E-Convex Programming                          | 1999 DOI<br>10.1023/a:1021792726715<br>(open)                |
| [@lasserre2014erratum]      | Erratum to: On convex optimization without convex representation                     | 2014 DOI 10.1007/s11590-014-0735-9 (open)                    |
| [@anon2004unconstrained]    | Unconstrained minimization                                                           | 2004 DOI<br>10.1017/cbo9780511804441.010<br>(open)           |
| [@yang2001econvex]          | On E-Convex Sets, E-Convex Functions, and E-Convex Programming                       | 2001 DOI<br>10.1023/a:1017532225395<br>(open)                |
| [@anon2004statistical]      | Statistical estimation                                                               | 2004 DOI<br>10.1017/cbo9780511804441.008<br>(open)           |
| [@anon2004mathematical]     | Mathematical background                                                              | 2004 DOI<br>10.1017/cbo9780511804441.013<br>(open)           |

# convex optimization VII

|                             |                                                                       |                                                         |
|-----------------------------|-----------------------------------------------------------------------|---------------------------------------------------------|
| [@anon2004geometric]        | Geometric problems                                                    | 2004 DOI<br>10.1017/cbo9780511804441.009<br>(open)      |
| [@anonduality]              | Duality Theory:<br>Monoduality in Convex<br>Optimization              | n.d. DOI 10.1007/springerref-<br>erence_72219 (open)    |
| [@anon2025coneconvex]       | Cone-Convex Functions:<br>Elementary Calculus and<br>Examples         | 2025 DOI<br>10.1017/9781009510561.025<br>(open)         |
| [@floudas1995convex]        | Convex Analysis                                                       | 1995 DOI<br>10.1093/oso/9780195100563.003.000<br>(open) |
| [@rubsamen2009robust]       | Robust broadband<br>adaptive beamforming<br>using convex optimization | 2009 DOI<br>10.1017/cbo9780511804458.010<br>(open)      |
| [@florezano2001convex]      | Convex Optimization<br>With Convex Constraints                        | 2001 DOI 10.1007/978-3-642-<br>56522-9_7 (open)         |
| [@anon2021generalizations]  | Generalizations of Convex<br>Functions                                | 2021 DOI<br>10.1002/9781119804093.ch4<br>(open)         |
| [@anon20144]                | 4 Convex Nonsmooth<br>Optimization                                    | 2014 DOI<br>10.2478/9783110426045.4<br>(open)           |
| [@anon2025convexd]          | Convex Optimization                                                   | 2025 DOI<br>10.1017/9781009493512.006<br>(open)         |
| [@anon2025separation]       | Separation Theorem and<br>Geometry of Convex Sets                     | 2025 DOI<br>10.1017/9781009510561.009<br>(open)         |
| [@dragomirescu1992smallest] | The smallest convex<br>extensions of a convex<br>function             | 1992 DOI<br>10.1080/02331939208843789<br>(open)         |
| [@anon2011algorithms]       | Algorithms for Conic<br>Optimization                                  | 2011 DOI 10.1201/b10839-12<br>(open)                    |

# convex optimization VIII

|                          |                                                                                                       |                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| [@anon2025convexe]       | Convex Optimization Theory                                                                            | 2025 DOI 10.1017/9781009428095.021 (open)       |
| [@strongin2000global]    | Global Optimization under Non-Convex Constraints — The Index Approach                                 | 2000 DOI 10.1007/978-1-4615-4677-1_6 (open)     |
| [@murota2024lconvex]     | L-Convex Functions and M-Convex Functions                                                             | 2024 DOI 10.1007/978-3-030-54621-2_325-1 (open) |
| [@anon2004equality]      | Equality constrained minimization                                                                     | 2004 DOI 10.1017/cbo9780511804441.011 (open)    |
| [@anon2004approximation] | Approximation and fitting                                                                             | 2004 DOI 10.1017/cbo9780511804441.007 (open)    |
| [@murota2008lconvex]     | L-convex Functions and M-convex Functions                                                             | 2008 DOI 10.1007/978-0-387-74759-0_325 (open)   |
| [@zalinescu2024locally]  | On locally Lipschitz convex functions defined on subsets with empty interior of locally convex spaces | 2024 DOI 10.1080/02331934.2024.2388200 (open)   |
| [@kramerbasis]           | Basis identification through convex optimization                                                      | n.d. DOI 10.31274/etd-180810-1098 (open)        |
| [@murota2001lconvex]     | L-Convex Functions and M-Convex Functions                                                             | 2001 DOI 10.1007/0-306-48332-7_244 (open)       |
| [@anon2004duality]       | Duality                                                                                               | 2004 DOI 10.1017/cbo9780511804441.006 (open)    |
| [@anonintroduction]      | Introduction                                                                                          | n.d. DOI 10.1017/cbo9781139924672.002 (open)    |
| [@anonalgorithm]         | Algorithm 1 : Convex optimization with regularized feature selection                                  | n.d. DOI 10.7717/peerj-cs.3752/table-101 (open) |

# convex optimization IX

|                               |                                                                      |                                              |
|-------------------------------|----------------------------------------------------------------------|----------------------------------------------|
| [@liuautonomous]              | Autonomous Trajectory Planning by Convex Optimization                | n.d. DOI 10.31274/etd-180810-3525 (open)     |
| [@anonbackground]             | Background                                                           | n.d. DOI 10.1017/cbo9781139924672.003 (open) |
| [@anonpreface]                | Preface                                                              | n.d. DOI 10.1017/cbo9781139924672.001 (open) |
| [@oktem2017computation al]    | Computational Spectral and Ultrafast Imaging via Convex Optimization | 2017 DOI 10.1007/978-3-319-61609-4_5 (open)  |
| [@anoneconomics]              | Economics                                                            | n.d. DOI 10.1017/cbo9781139924672.007 (open) |
| [@anon2011cones]              | Cones, Complementarity, and Conic Optimization                       | 2011 DOI 10.1201/b10839-5 (open)             |
| [@zaslavski2020minimizati on] | Minimization of Sharp Weakly Convex Functions                        | 2020 DOI 10.1007/978-3-030-37822-6_11 (open) |

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# stochastic gradient descent I

*Papers retrieved: 100 · per-source contributions: arxiv=100,  
crossref=100 · backend errors: none*

| Cite                                | Title                                                                                             | Year DOI / URL                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------|
| [@gwozdz2018stochastic]             | Stochastic and semi-stochastic gradient descent in speech disambiguation                          | 2018 DOI 10.7490/f1000research.1115776.1 (open)    |
| [@wang2018stochastic]               | Stochastic gradient descent                                                                       | 2018 DOI 10.53347/rid-61715 (open)                 |
| [@kosinski2017coxphsgd]             | coxphSGD: Stochastic Gradient Descent log-Likelihood Estimation in Cox Proportional Hazards Model | 2017 DOI 10.32614/cran.package.coxphsgd (open)     |
| [@anontable]                        | Table 4: Stochastic gradient descent (SGD) parameter settings with CC images.                     | n.d. DOI 10.7717/peerj-cs.3332/table-4 (open)      |
| [@peseux2023stochastic]             | Stochastic Gradient Descent with Gradient Estimator for Categorical Features                      | 2023 DOI 10.2139/ssrn.4439301 (open)               |
| [@theodoridis2015stochastic]        | Stochastic Gradient Descent                                                                       | 2015 DOI 10.1016/b978-0-12-801522-3.00005-7 (open) |
| [@anon2014stochastic]               | Stochastic Gradient Descent                                                                       | 2014 DOI 10.1017/cbo9781107298019.015 (open)       |
| [@anon1011171223051015783306294001] | 10.1117/12.2305101.5783306294001                                                                  | n.d. DOI 10.1117/12.2305101.5783306294001 (open)   |

# stochastic gradient descent II

|                                     |                                                                                                        |                                                   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| [@lan2023stochastic]                | Stochastic Gradient Descent                                                                            | 2023 DOI 10.1007/978-3-030-54621-2_777-1 (open)   |
| [@anon1011171225128176013939792001] | 10.1117/12.2512817.6013939792001                                                                       | n.d. DOI 10.1117/12.2512817.6013939792001 (open)  |
| [@anon1011171226435646314793833112] | 10.1117/12.2643564.6314793833112                                                                       | n.d. DOI 10.1117/12.2643564.6314793833112 (open)  |
| [@songanglebased]                   | An Angle-based Stochastic Gradient Descent Method for Machine Learning: Principle and Application      | n.d. DOI 10.25148/etd.fidc009555 (open)           |
| [@turali2024optimal]                | Optimal Stochastic Gradient Descent                                                                    | 2024 DOI 10.2139/ssrn.4879640 (open)              |
| [@ketkar2017stochastic]             | Algorithm for Filtering Stochastic Gradient Descent                                                    | 2017 DOI 10.1007/978-1-4842-2766-4_8 (open)       |
| [@alonso2025mathematics]            | The Mathematics of Stochastic Gradient Descent and Non-Convex Optimization                             | 2025 DOI 10.2139/ssrn.5032378 (open)              |
| [@pillaudvivienlearning]            | Learning with reproducing kernel Hilbert spaces : stochastic gradient descent and laplacian estimation | n.d. DOI 10.70675/f938eaabz37fcz4483zba85z (open) |
| [@amor2024realtime]                 | Real-Time Traffic Prediction Through Stochastic Gradient Descent                                       | 2024 DOI 10.5220/0012687400003702 (open)          |
| [@yang2022adaptive]                 | Adaptive stochastic gradient descent for large-scale learning problems                                 | 2022 DOI 10.21203/rs.3.rs-1066512/v1 (open)       |

# stochastic gradient descent III

|                                                        |                                                                                                                                    |                                                                            |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| [@arefin2016minimizing]                                | Minimizing Average of Loss Functions Using Gradient Descent and Stochastic Gradient Descent                                        | 2016 DOI<br>10.3329/dujs.v64i2.54490<br>(open)                             |
| [@chendistributed]                                     | Distributed Stochastic Gradient Descent with Staleness: A Stochastic Delay Differential Equation Based Framework_supp1-3546574.pdf | n.d. DOI<br>10.1109/tsp.2025.3546574/mm1<br>(open)                         |
| [@abubaker2022scaling]                                 | Scaling Stratified Stochastic Gradient Descent for Distributed Matrix Completion                                                   | 2022 DOI<br>10.36227/techrxiv.19350536.v1<br>(open)                        |
| [@fu2025theoretical]                                   | A Theoretical and Experimental Study of Gradient Descent and Its Stochastic Variants                                               | 2025 DOI<br>10.36227/techrxiv.176162233.330760<br>(open)                   |
| [@abubaker2022scalinga]                                | Scaling Stratified Stochastic Gradient Descent for Distributed Matrix Completion                                                   | 2022 DOI<br>10.36227/techrxiv.19350536<br>(open)                           |
| [@anon101117123028105e628af96d7c5ee11a99e00505691c5e1] | 10.1117/12.3028105.e628af96-d7c5-ee11-a99e-00505691c5e1                                                                            | n.d. DOI<br>10.1117/12.3028105.e628af96-d7c5-ee11-a99e-00505691c5e1 (open) |
| [@tarkhan2020bigSurvsgd]                               | bigSurvSGD: Big Survival Analysis Using Stochastic Gradient Descent                                                                | 2020 DOI<br>10.32614/cran.package.bigSurvsgd<br>(open)                     |
| [@lam2023resampling]                                   | Resampling Stochastic Gradient Descent Cheaply                                                                                     | 2023 DOI<br>10.1109/wsc60868.2023.10408023<br>(open)                       |

# stochastic gradient descent IV

|                            |                                                                                                                                        |                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| [@sun2023noisy]            | Noisy Stochastic Gradient Descent Algorithm with Stochastic Event-Triggered Mechanism for Communication-Efficient Distributed Learning | 2023 DOI 10.2139/ssrn.4588764 (open)               |
| [@theodoridis2020online]   | Online Learning: the Stochastic Gradient Descent Family of Algorithms                                                                  | 2020 DOI 10.1016/b978-0-12-818803-3.00014-3 (open) |
| [@theodoridis2026online]   | Online learning: the stochastic gradient descent family of algorithms                                                                  | 2026 DOI 10.1016/b978-0-44-329238-5.00011-1 (open) |
| [@ali2023comparing]        | Comparing the Effectiveness of Support Vector Classifier and Stochastic Gradient Descent in Hate-Speech Detection                      | 2023 DOI 10.47611/harp.315 (open)                  |
| [@guo2022variable]         | Variable selection of regularized stochastic gradient descent in logistic regression                                                   | 2022 DOI 10.54647/mathematics11319 (open)          |
| [@cacciola2023convergence] | On the Convergence of Stochastic Gradient Descent in Low-Precision Number Formats                                                      | 2023 DOI 10.5220/0011795500003411 (open)           |
| [@sirignano2017stochastic] | Stochastic Gradient Descent in Continuous Time                                                                                         | 2017 DOI 10.2139/ssrn.2954149 (open)               |
| [@abrolbekov2026high]      | High Probability Convergence Bounds for Non-convex Stochastic Gradient Descent with                                                    | 2026 DOI 10.2139/ssrn.6738058 (open)               |



# stochastic gradient descent V

[@anon101117122599744 10.1117/12.2599744.6269411057001 6269411057001]

n.d. DOI

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(open)

[@nguyenheavytailed] Heavy-tailed nature of stochastic gradient descent in deep learning : theoretical and empirical analysis

n.d. DOI

10.70675/674d4873zb009z425cz9c8e  
(open)

[@shrimali2023comparative] A comparative study of gradient descent and stochastic gradient descent method for optimization

2023 DOI 10.1063/5.0148736  
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[@anon2011largescale] Large-Scale Machine Learning with Stochastic Gradient Descent Léon Bottou

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(open)

[@anon2021stochastic] Stochastic Gradient Descent

2021 DOI

10.1017/9781108860604.006  
(open)

[@yamada2018hyperparameterfree] Hyperparameter-free optimizer of stochastic gradient descent that incorporates unit correction and moment estimation

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[@li2024stochastic] Stochastic Gradient Descent in Nonconvex Optimization: Continuous-Time Dynamics and the Role of Learning Rates

2024 DOI

10.2139/ssrn.4980990  
(open)

[@cai2023communicationefficient] Communication-Efficient Distributed Stochastic Gradient Descent with Pooling Operator

2023 DOI

10.2139/ssrn.4327869  
(open)

# stochastic gradient descent VI

|                              |                                                                                                                            |                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| [@ayadi2021stochastic]       | Stochastic Runge-Kutta methods and adaptive SGD-G2 stochastic gradient descent                                             | 2021 DOI<br>10.1109/icpr48806.2021.9412831<br>(open) |
| [@chen2026fractionalorder]   | Fractional-order dynamics driven stochastic gradient descent method with momentum                                          | 2026 DOI<br>10.2139/ssrn.6902318<br>(open)           |
| [@srinivasan2026intelligent] | An Intelligent Predictive CPU Scheduling Algorithm Using Online Multiple Linear Regression and Stochastic Gradient Descent | 2026 DOI<br>10.2139/ssrn.6956552<br>(open)           |
| [@kour2024stochastic]        | Stochastic Gradient Descent Optimized Efficient Transfer Learning Architecture for Brain Tumor Segmentation                | 2024 DOI<br>10.2139/ssrn.5012382<br>(open)           |
| [@ramos2015hilbert]          | Hilbert maps: scalable continuous occupancy mapping with stochastic gradient descent                                       | 2015 DOI<br>10.15607/rss.2015.xi.002<br>(open)       |
| [@sirignano2020stochastic]   | Stochastic Gradient Descent in Continuous Time: A Central Limit Theorem                                                    | 2020 DOI<br>10.1287/stsy.2019.0050<br>(open)         |
| [@hafshejani2023fast]        | Fast Armijo line search for stochastic gradient descent                                                                    | 2023 DOI 10.21203/rs.3.rs-2285238/v1 (open)          |
| [@bottou2010largescale]      | Large-Scale Machine Learning with Stochastic Gradient Descent                                                              | 2010 DOI 10.1007/978-3-7908-2604-3_16 (open)         |

# stochastic gradient descent VII

|                             |                                                                                                             |                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| [@chen2025revisit]          | Revisit Stochastic Gradient Descent for Strongly Convex Objectives: Tight Uniform-in-Time Bounds            | 2025 DOI<br>10.2139/ssrn.5546711<br>(open)            |
| [@cheng2025convergence]     | Convergence Analysis of the Last Iterate in Distributed Stochastic Gradient Descent with Momentum           | 2025 DOI<br>10.2139/ssrn.5351413<br>(open)            |
| [@huadaptive]               | Adaptive Batch Size Time Evolving Stochastic Gradient Descent for Federated Learning_supp1-3610169.pdf      | n.d. DOI<br>10.1109/tpami.2025.3610169/mm1<br>(open)  |
| [@amari1993backpropagation] | Backpropagation and stochastic gradient descent method                                                      | 1993 DOI 10.1016/0925-2312(93)90006-o (open)          |
| [@schraudolph1999local]     | Local gain adaptation in stochastic gradient descent                                                        | 1999 DOI<br>10.1049/cp:19991170<br>(open)             |
| [@anonymous2025noise]       | Noise balance and stationary distribution of stochastic gradient descent                                    | 2025 DOI 10.1103/zjq6-nzwd<br>(open)                  |
| [@liu2024optimizing]        | Optimizing Synchronous Stochastic Gradient Descent with Local Efficient Sign and Model Averaging Correction | 2024 DOI<br>10.2139/ssrn.4965637<br>(open)            |
| [@ito2026constrained]       | Constrained Graph Drawing by Stochastic Gradient Descent                                                    | 2026 DOI 10.1109/pacificvis68791.2026.00006<br>(open) |

# stochastic gradient descent VIII

|                              |                                                                                                                                                          |                                                 |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| [@chang2020efficient]        | Efficient phase-locking of<br>60 fiber lasers by<br>stochastic parallel<br>gradient descent<br>algorithm                                                 | 2020 DOI<br>10.3788/col202018.101403<br>(open)  |
| [@bao2020stochastic]         | Stochastic gradient<br>descent algorithm for<br>stochastic optimization in<br>solving analytic<br>continuation problems                                  | 2020 DOI<br>10.3934/fods.2020001<br>(open)      |
| [@li2024withdrawn]           | WITHDRAWN: Bridging<br>Stochastic Gradient<br>Descent and Markov<br>Chains: Constant<br>Step-Size Convergence<br>and Richardson-Romberg<br>Extrapolation | 2024 DOI 10.21203/rs.3.rs-<br>5202271/v1 (open) |
| [@livni2024sample]           | The Sample Complexity<br>of Gradient Descent in<br>Stochastic Convex<br>Optimization                                                                     | 2024 DOI<br>10.52202/079017-2048<br>(open)      |
| [@li2024withdrawna]          | WITHDRAWN: Bridging<br>Stochastic Gradient<br>Descent and Markov<br>Chains: Constant<br>Step-Size Convergence<br>and Richardson-Romberg<br>Extrapolation | 2024 DOI 10.21203/rs.3.rs-<br>5202271/v2 (open) |
| [@koutsibella2020stochastic] | Stochastic gradient<br>descent possibilistic<br>clustering                                                                                               | 2020 DOI<br>10.1145/3411408.3411436<br>(open)   |
| [@chen2012dictionary]        | Dictionary learning with<br>weighted stochastic<br>gradient descent                                                                                      | 2012 DOI 10.1109/ic-<br>cps.2012.6384229 (open) |

# stochastic gradient descent IX

|                                         |                                                                                                                                                          |                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| [@t2024hybrid]                          | Hybrid Optimization<br>Model Integrating<br>Gradient Descent and<br>Stochastic Descent for<br>Enhanced Osteoporosis<br>and Osteopenia<br>Recognition     | 2024 DOI<br>10.53759/7669/jmc202404032<br>(open)       |
| [@lee2017differentially]                | Differentially Private<br>Variance Reduced<br>Stochastic Gradient<br>Descent                                                                             | 2017 DOI<br>10.1109/ictcs.2017.60<br>(open)            |
| [@anon101117122525953<br>6062680904001] | 10.1117/12.2525953.6062680904001                                                                                                                         | n.d. DOI<br>10.1117/12.2525953.6062680904001<br>(open) |
| [@ravasi2022multidimensi<br>onal]       | Multi-Dimensional<br>Deconvolution with<br>Stochastic Gradient<br>Descent                                                                                | 2022 DOI 10.3997/2214-<br>4609.202210234 (open)        |
| [@hu2022efficiency]                     | Efficiency Ordering of<br>Stochastic Gradient<br>Descent                                                                                                 | 2022 DOI<br>10.52202/068431-1155<br>(open)             |
| [@azimjonov2023stochasti<br>c]          | Stochastic Gradient<br>Descent Classifier-Based<br>Lightweight Intrusion<br>Detection Systems Using<br>the Most Efficient Feature<br>Subsets of Datasets | 2023 DOI<br>10.2139/ssrn.4378339<br>(open)             |
| [@sharma2021guided]                     | Guided parallelized<br>stochastic gradient<br>descent for delay<br>compensation                                                                          | 2021 DOI<br>10.1016/j.j.asoc.2021.107084<br>(open)     |
| [@yaghoubi2017hybrid]                   | Hybrid approximate<br>gradient and stochastic<br>descent for falsification of<br>nonlinear systems                                                       | 2017 DOI<br>10.23919/acc.2017.7963007<br>(open)        |

# stochastic gradient descent X

|                                                        |                                                                                                            |                                                                            |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| [@data2020byzantineresilient]                          | On Byzantine-Resilient High-Dimensional Stochastic Gradient Descent                                        | 2020 DOI<br>10.1109/isit44484.2020.9174363 (open)                          |
| [@cheng2019static]                                     | Static & Dynamic Appointment Scheduling with Stochastic Gradient Descent                                   | 2019 DOI<br>10.23919/acc.2019.8814666 (open)                               |
| [@anon1011171230261530df877b5f9b8ee11a99dc49c781f4d15] | 10.1117/12.3026153.0df877b5-f9b8-ee11-a99d-c49c781f4d15                                                    | n.d. DOI<br>10.1117/12.3026153.0df877b5-f9b8-ee11-a99d-c49c781f4d15 (open) |
| [@song2021agsgd]                                       | AG-SGD: Angle-Based Stochastic Gradient Descent                                                            | 2021 DOI 10.1109/access.2021.3055993 (open)                                |
| [@jain2025controlling]                                 | Controlling stochastic gradient descent using stochastic approximation for robust distributed optimization | 2025 DOI<br>10.3934/naco.2024041 (open)                                    |
| [@williams2022stochastic]                              | Stochastic Gradient Descent for Optimization of Nuclear Systems                                            | 2022 DOI 10.21203/rs.3.rs-2073277/v1 (open)                                |
| [@anon2014convergence]                                 | On the Convergence Rate of Stochastic Gradient Descent for Strongly Convex Functions                       | 2014 DOI 10.1201/b17558-10 (open)                                          |
| [@zamora2016dendrite]                                  | Dendrite morphological neurons trained by stochastic gradient descent                                      | 2016 DOI<br>10.1109/ssci.2016.7849933 (open)                               |
| [@liu2021diffusion]                                    | A Diffusion Approximation Theory of Momentum Stochastic Gradient Descent in Nonconvex Optimization         | 2021 DOI<br>10.1287/stsy.2021.0083 (open)                                  |

# stochastic gradient descent XI

|                               |                                                                                                                                            |                                                       |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| [@hua2023machinelearnin<br>g] | Machine-Learning<br>Topology Optimization<br>with Stochastic Gradient<br>Descent Optimizer for<br>Heat Conduction<br>Problems              | 2023 DOI<br>10.2139/ssrn.4594476<br>(open)            |
| [@sundecentralized]           | On the Decentralized<br>Stochastic Gradient<br>Descent with Markov<br>Chain Sampling_supp1-<br>3297053.pdf                                 | n.d. DOI<br>10.1109/tsp.2023.3297053/mm1<br>(open)    |
| [@bhardwaj2016practical]      | Practical Efficiency of<br>Asynchronous Stochastic<br>Gradient Descent                                                                     | 2016 DOI<br>10.1109/mlhpc.2016.010<br>(open)          |
| [@iutzeler2024derivatives]    | Derivatives of Stochastic<br>Gradient Descent in<br>parametric optimization                                                                | 2024 DOI<br>10.52202/079017-3775<br>(open)            |
| [@jiao2024emergence]          | Emergence of heavy tails<br>in homogenized stochastic<br>gradient descent                                                                  | 2024 DOI<br>10.52202/079017-0450<br>(open)            |
| [@cui2019acoustic]            | Acoustic Model<br>Optimization Based on<br>Evolutionary Stochastic<br>Gradient Descent with<br>Anchors for Automatic<br>Speech Recognition | 2019 DOI<br>10.21437/interspeech.2019-<br>2620 (open) |
| [@ferrarotti2019synthesis]    | Synthesis of Optimal<br>Feedback Controllers<br>from Data via Stochastic<br>Gradient Descent                                               | 2019 DOI<br>10.23919/ecc.2019.8796130<br>(open)       |
| [@wang2014robust]             | Robust pose graph<br>optimization using<br>stochastic gradient<br>descent                                                                  | 2014 DOI<br>10.1109/icra.2014.6907482<br>(open)       |

# stochastic gradient descent XII

|                               |                                                                                                          |                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| [@wijnhoven2010fast]          | Fast Training of Object Detection Using Stochastic Gradient Descent                                      | 2010 DOI<br>10.1109/icpr.2010.112<br>(open)            |
| [@sharma2018guided]           | Guided Stochastic Gradient Descent Algorithm for inconsistent datasets                                   | 2018 DOI<br>10.1016/j.asoc.2018.09.038<br>(open)       |
| [@anon2020testing]            | Testing of the stochastic parallel gradient descent algorithm to the alignment of a two-mirror telescope | 2020 DOI 10.17586/1023-5086-2020-87-05-31-41<br>(open) |
| [@koren2022benign]            | Benign Underfitting of Stochastic Gradient Descent                                                       | 2022 DOI<br>10.52202/068431-1425<br>(open)             |
| [@afshar2009gradient]         | Gradient descent optimisation for ILC-based stochastic distribution control                              | 2009 DOI<br>10.1109/icca.2009.5410612<br>(open)        |
| [@anon2025neural]             | Neural Networks, Backpropagation, and Stochastic Gradient Descent                                        | 2025 DOI<br>10.1017/9781009509435.009<br>(open)        |
| [@li2021fast]                 | Fast Distributed Stochastic Nesterov Gradient Descent Algorithm for Image Classification                 | 2021 DOI<br>10.1109/cac53003.2021.9727635<br>(open)    |
| [@mukhopadhyay2020stochastic] | Stochastic gradient descent for linear systems with sequential matrix entry accumulation                 | 2020 DOI<br>10.1016/j.sigpro.2020.107494<br>(open)     |
| [@christensen2024learning]    | Is Learning in Biological Neural Networks Based on Stochastic Gradient Descent?                          | 2024 DOI<br>10.1162/neco_a_01668<br>(open)             |



# stochastic gradient descent XIII

[@archibald2020stochastic A Stochastic Gradient  
] Descent Approach for  
Stochastic Optimal  
Control

2020 DOI 10.4208/ea-  
jam.190420.200420  
(open)

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# reproducible research I

*Papers retrieved: 100 · per-source contributions: arxiv=100,  
crossref=100 · backend errors: none*

| Cite                             | Title                                                      | Year DOI / URL                                        |
|----------------------------------|------------------------------------------------------------|-------------------------------------------------------|
| [@gandrud2018introduc<br>g]      | Introducing Reproducible Research                          | 2018 DOI<br>10.1201/9781315382548-1 (open)            |
| [@gandrud2020introduc<br>g]      | Introducing Reproducible Research                          | 2020 DOI<br>10.1201/9780429031854-2 (open)            |
| [@gandrud2018getting]            | Getting Started with Reproducible Research                 | 2018 DOI<br>10.1201/9781315382548-2 (open)            |
| [@gandrud2020getting]            | Getting Started with Reproducible Research                 | 2020 DOI<br>10.1201/9780429031854-3 (open)            |
| [@hoeftling2018reproducib<br>le] | Reproducible Research for Large-Scale Data Analysis        | 2018 DOI<br>10.1201/9781315373461-8 (open)            |
| [@xie2018knitr]                  | knitr: A Comprehensive Tool for Reproducible Research in R | 2018 DOI<br>10.1201/9781315373461-1 (open)            |
| [@baker2022reproducible]         | Reproducible data analysis                                 | 2022 DOI<br>10.1093/hesc/9780192896599.003.001 (open) |
| [@anoninbioinformatics]          | NinBioinformatics Reproducible Research Reports            | n.d. DOI 10.21428/3c290898 (open)                     |
| [@anon2019reproducible]          | Reproducible Workflow                                      | 2019 DOI<br>10.2307/j.ctvpb3xkg.16 (open)             |

# reproducible research II

|                                   |                                                                                                                        |                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| [@preeyanon2018reproduc<br>ible]  | Reproducible<br>Bioinformatics Research<br>for Biologists                                                              | 2018 DOI<br>10.1201/9781315373461-<br>7 (open)                     |
| [@davison2018sumatra]             | Sumatra: A Toolkit for<br>Reproducible Research                                                                        | 2018 DOI<br>10.1201/9781315373461-<br>3 (open)                     |
| [@anon2019eleven]                 | ELEVEN. Reproducible<br>Workflow                                                                                       | 2019 DOI<br>10.1525/9780520969230-<br>014 (open)                   |
| [@hrynaskiewicz2018ope<br>n]      | Open Science and the<br>Role of Publishers in<br>Reproducible Research                                                 | 2018 DOI<br>10.1201/9781315373461-<br>15 (open)                    |
| [@anonsupplemental]               | Supplemental Information<br>1: Reproducible Research<br>Instructions.                                                  | n.d. DOI 10.7717/peerj-<br>cs.904/supp-1 (open)                    |
| [@stodden2014implementi<br>ng]    | Implementing<br>Reproducible Research                                                                                  | 2014 DOI 10.1201/b16868<br>(open)                                  |
| [@anonreproducible]               | Reproducible Research<br>Data and Software                                                                             | n.d. DOI<br>10.12688/f1000research.channels.908<br>(open)          |
| [@kedron2023reproducible<br>]     | Reproducible Research<br>Practices and Barriers to<br>Reproducible Research in<br>Geography: Insights from<br>a Survey | 2023 DOI<br>10.31219/osf.io/nyrq9<br>(open)                        |
| [@murrayrust2018reprodu<br>cible] | Reproducible Physical<br>Science and the<br>Declaratron                                                                | 2018 DOI<br>10.1201/9781315373461-<br>5 (open)                     |
| [@burgess2018reproducibl<br>e]    | Reproducible research<br>article collection                                                                            | 2018 DOI 10.14293/s2199-<br>1006.1.sor-<br>uncat.clsuuhc.v1 (open) |
| [@solt2016pewdata]                | pewdata: Reproducible<br>Retrieval of Pew Research<br>Center Datasets                                                  | 2016 DOI<br>10.32614/cran.package.pewdata<br>(open)                |

# reproducible research III

[@basu2017reproducible] Reproducible research  
with jupyter notebooks

[@edmunds2016reproducible] Reproducible Research  
Resources for  
Research(ing) Parasites

[@edmunds2016reproducible] Reproducible Research  
Resources for  
Research(ing) Parasites

[@hector2021reproducible]  
Reproducible Research

[@edmunds2015fermenting] Fermenting a  
Reproducible Research  
Revolution

[@kunisato2019introduction] Introduction to  
reproducible psychological  
research

[@suber2008oa] OA for reproducible  
research

[@hinsen2013platforms] Platforms for reproducible  
research

[@edmunds2015fermenting] Fermenting a  
Reproducible Research  
Revolution

[@turek2019case] Case Studies in  
Reproducible Research

[@gandrud2020reproducible] Reproducible Research  
with R and RStudio

[@basu2018reproducible] Reproducible research  
using minimalist plain  
text tools

2017 DOI  
10.22541/au.151460905.57485984  
(open)

2016 DOI  
10.59350/63nv3-fa097  
(open)

2016 DOI  
10.59350/vhpgg-ec668  
(open)

2021 DOI  
10.1093/oso/9780198798170.003.000  
(open)

2015 DOI  
10.59350/ejn9k-s7c54  
(open)

2019 DOI  
10.31234/osf.io/x8js5  
(open)

2008 DOI  
10.63485/4jjac-1dd97  
(open)

2013 DOI  
10.59350/s56jr-3cn95  
(open)

2015 DOI  
10.59350/0hs8b-m2239  
(open)

2019 DOI  
10.1525/9780520967779-  
006 (open)

2020 DOI  
10.1201/9780429031854  
(open)

2018 DOI 10.32388/649864  
(open)

# reproducible research IV

|                                   |                                                             |                                                       |
|-----------------------------------|-------------------------------------------------------------|-------------------------------------------------------|
| [@charlton2016how]                | How do we ensure that research is reproducible?             | 2016 DOI<br>10.15200/winn.146397.78741<br>(open)      |
| [@gandrud2018reproducib<br>le]    | Reproducible Research with R and RStudio                    | 2018 DOI<br>10.1201/9781315382548<br>(open)           |
| [@bahlai2016open]                 | The open science and reproducible research course           | 2016 DOI<br>10.7490/f1000research.1112876.1<br>(open) |
| [@gandrud2013reproducib<br>le]    | Reproducible Research with R and R Studio                   | 2013 DOI 10.1201/b15100<br>(open)                     |
| [@edmunds2014carmen]              | CARMEN, reproducible research and push-button papers        | 2014 DOI<br>10.59350/xx4rg-zcd90<br>(open)            |
| [@kitzes2019basic]                | The Basic Reproducible Workflow Template                    | 2019 DOI<br>10.1525/9780520967779-005 (open)          |
| [@missingvalueplatforms]          | Platforms                                                   | n.d. DOI 10.1201/b16868-20<br>(open)                  |
| [@white2018software]              | Software skills for reproducible data-intensive research    | 2018 DOI<br>10.7490/f1000research.1115901.1<br>(open) |
| [@weisbrod2026examplepr<br>oject] | example-project: A Reproducible Empirical Research Template | 2026 DOI<br>10.31235/osf.io/yx7af_v1<br>(open)        |
| [@anon2020transparent]            | Transparent and Reproducible Data Analysis                  | 2020 DOI<br>10.4135/9781526421036926480<br>(open)     |
| [@suber2008open]                  | Open licensing to enable reproducible research              | 2008 DOI<br>10.63485/8k73p-vrh09<br>(open)            |
| [@anon2016reproducible]           | Reproducible Research with R and R Studio, Second Edition   | 2016 DOI 10.1201/b18546<br>(open)                     |

# reproducible research V

|                           |                                                                                 |                                                     |
|---------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------|
| [@missingvaluetools]      | Tools                                                                           | n.d. DOI 10.1201/b16868-6<br>(open)                 |
| [@edmunds2014carmena]     | CARMEN, reproducible research and push-button papers                            | 2014 DOI 10.59350/p6xyp-h2p77<br>(open)             |
| [@anonfigure]             | Figure 2: The final reproducible research criteria used for the evaluation.     | n.d. DOI 10.7717/peerj.5072/fig-2<br>(open)         |
| [@anon2025preparing]      | Preparing for Transparent and Reproducible Quantitative Social Science Research | 2025 DOI 10.4135/9781036230647<br>(open)            |
| [@hinsen2016reproducible] | From reproducible to verifiable computer-aided research                         | 2016 DOI 10.59350/fj0g3-5yv59<br>(open)             |
| [@suber2009more]          | More on OA to facilitate reproducible research                                  | 2009 DOI 10.63485/tys9t-3y862<br>(open)             |
| [@anon1061686udeac59432]  | 10.61686/udeac59432                                                             | n.d. DOI 10.61686/udeac59432<br>(open)              |
| [@geertreproducible]      | Reproducible research reports with Quarto                                       | n.d. DOI 10.21428/1192f2f8.fb1a1b7f<br>(open)       |
| [@anon2016reproduciblea]  | Reproducible Research                                                           | 2016 DOI 10.1201/b15166-10<br>(open)                |
| [@anon2018implementing]   | Implementing Reproducible Research                                              | 2018 DOI 10.1201/9781315373461<br>(open)            |
| [@heston2023statistics]   | Statistics, Ethics, and the Promotion of Reproducible Research                  | 2023 DOI 10.22541/au.169627960.06669688/v<br>(open) |
| [@moresi2018alaska]       | Alaska Moho Model (Reproducible research with containers)                       | 2018 DOI 10.59350/pn8gh-98592<br>(open)             |

# reproducible research VI

|                                  |                                                                                                |
|----------------------------------|------------------------------------------------------------------------------------------------|
| [@hinsen2013python]              | Python as a platform for reproducible research                                                 |
| [@wiebels2021leveraging]         | Leveraging Containers for Reproducible Psychological Research                                  |
| [@marwick2019case]               | Case Study 12: Using R and Related Tools for Reproducible Research in Archaeology              |
| [@roskar2022renku]               | Renku: a platform for collaborative, reusable and reproducible research                        |
| [@gandrud2020conclusion]         | Conclusion                                                                                     |
| [@maerz2024quarto]               | Quarto in RStudio: Writing Reproducible & Dynamic Research Papers                              |
| [@anon2021decision]              | Decision letter: Is preclinical research in cancer biology reproducible enough?                |
| [@hediyehzadeh2016computational] | Computational workflows for research students: towards a reproducible research                 |
| [@gandrud2018conclusion]         | Conclusion                                                                                     |
| [@missingvaluepractices]         | Practices and Guidelines                                                                       |
| [@limarereproducible]            | Reproducible research, software quality, online interfaces and publishing for image processing |

|          |                                          |
|----------|------------------------------------------|
| 2013 DOI | 10.59350/85pnj-bak53 (open)              |
| 2021 DOI | 10.31234/osf.io/h7tkg (open)             |
| 2019 DOI | 10.1525/9780520967779-021 (open)         |
| 2022 DOI | 10.5194/egusphere-egu22-10697 (open)     |
| 2020 DOI | 10.1201/9780429031854-17 (open)          |
| 2024 DOI | 10.61700/pdeaaefq62n681556 (open)        |
| 2021 DOI | 10.7554/elife.67527.sa1 (open)           |
| 2016 DOI | 10.7490/f1000research.1113332.1 (open)   |
| 2018 DOI | 10.1201/9781315382548-14 (open)          |
| n.d. DOI | 10.1201/b16868-12 (open)                 |
| n.d. DOI | 10.70675/d17fee18z3990z46f0z8820z (open) |

# reproducible research VII

|                          |                                                                                                                        |                                                       |
|--------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| [@moresi2018alaskaa]     | Alaska Moho Model<br>(Reproducible research<br>with containers)                                                        | 2018 DOI<br>10.59350/15qd1-96r10<br>(open)            |
| [@boettiger2019case]     | Case Study 3: A<br>Reproducible R Notebook<br>Using Docker                                                             | 2019 DOI<br>10.1525/9780520967779-<br>012 (open)      |
| [@solt2016icpsrdata]     | icpsrdata: Reproducible<br>Data Retrieval from the<br>ICPSR Archive                                                    | 2016 DOI<br>10.32614/cran.package.icpsrdata<br>(open) |
| [@peyrache2026elife]     | eLife Assessment:<br>Spyglass: a framework for<br>reproducible and<br>shareable neuroscience<br>research               | 2026 DOI<br>10.7554/elife.108089.2.sa3<br>(open)      |
| [@anon2025reviewer]      | Reviewer #1 (Public<br>review): Spyglass: a<br>framework for<br>reproducible and<br>shareable neuroscience<br>research | 2025 DOI<br>10.7554/elife.108089.1.sa1<br>(open)      |
| [@gavel2025ensuring]     | Ensuring Your<br>Quantitative Research is<br>Replicable and<br>Reproducible                                            | 2025 DOI<br>10.4135/9781036216696<br>(open)           |
| [@hinsen2017sustainable] | Sustainable software and<br>reproducible research:<br>dealing with software<br>collapse                                | 2017 DOI<br>10.59350/7tavk-2hf75<br>(open)            |
| [@basu2021how]           | How to use orgmode for<br>reproducible research,<br>rough cut version 1                                                | 2021 DOI 10.32388/b5rvo7<br>(open)                    |
| [@turner2010sweave]      | Sweave for Reproducible<br>Research and Beautiful<br>Statistical Reports                                               | 2010 DOI<br>10.59350/247k5-kfv86<br>(open)            |



# reproducible research VIII

|                             |                                                                                                         |                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| [@turner2009seminar]        | Seminar: Reproducible Research with R, LaTeX, & Sweave                                                  | 2009 DOI<br>10.59350/2958f-vdz74 (open)       |
| [@peyrache2025elife]        | eLife Assessment: Spyglass: a framework for reproducible and shareable neuroscience research            | 2025 DOI<br>10.7554/elife.108089.1.sa2 (open) |
| [@lebeau2020reproducible]   | Reproducible Analyses in Educational Research                                                           | 2020 DOI 10.17077/pp.005637 (open)            |
| [@yildiz2023dataintegrated] | Data-integrated executable publications for reproducible geohazards research                            | 2023 DOI 10.5194/egusphere-egu23-7417 (open)  |
| [@anon2017fostering]        | Fostering reproducible fMRI research                                                                    | 2017 DOI<br>10.1038/ncomms14748 (open)        |
| [@anon2026reviewer]         | Reviewer #2 (Public review): Spyglass: a framework for reproducible and shareable neuroscience research | 2026 DOI<br>10.7554/elife.108089.2.sa1 (open) |
| [@anon2025reviewer]         | Reviewer #2 (Public review): Spyglass: a framework for reproducible and shareable neuroscience research | 2025 DOI<br>10.7554/elife.108089.1.sa0 (open) |
| [@chow2019reproducible]     | Reproducible Research                                                                                   | 2019 DOI<br>10.1201/9780429275067-9 (open)    |
| [@alston2020beginners]      | A Beginner's Guide to Conducting Reproducible Research                                                  | 2020 DOI<br>10.32942/osf.io/h5r6n (open)      |

# reproducible research IX

|                           |                                                                                                         |                                                          |
|---------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| [@hinsen2017reproducible] | Reproducible research in the Python ecosystem: a reality check                                          | 2017 DOI<br>10.59350/wgebd-68y68<br>(open)               |
| [@turner2010sweavea]      | Sweave for Reproducible Research and Beautiful Statistical Reports                                      | 2010 DOI<br>10.59350/pexqx-j5g86<br>(open)               |
| [@hastings2023ai]         | AI for Reproducible Research                                                                            | 2023 DOI<br>10.1201/9781003226642-4 (open)               |
| [@anon2026reviewera]      | Reviewer #1 (Public review): Spyglass: a framework for reproducible and shareable neuroscience research | 2026 DOI<br>10.7554/elife.108089.2.sa2<br>(open)         |
| [@strand2019publishing]   | Publishing open, reproducible research with undergraduates                                              | 2019 DOI<br>10.31234/osf.io/f7kuy<br>(open)              |
| [@hinsen2012unifying]     | Unifying version control and dependency management for reproducible research                            | 2012 DOI<br>10.59350/986d5-0he30<br>(open)               |
| [@mittal2025explainable]  | Explainable AI-Augmented DevSecOps for Secure and Reproducible Cloud-Native Research Software           | 2025 DOI<br>10.36227/techrxiv.175617187.787756<br>(open) |
| [@ohta2015containerbased] | Container-based sequence data analysis workflow for reproducible research                               | 2015 DOI<br>10.7490/f1000research.1110170.1<br>(open)    |
| [@turner2009seminara]     | Seminar: Reproducible Research with R, LaTeX, & Sweave                                                  | 2009 DOI<br>10.59350/d3cd0-5jy93<br>(open)               |

# reproducible research X

|                               |                                                                                           |                                                  |
|-------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------|
| [@butland2019community<br>]   | Community Call -<br>Reproducible Research<br>with R                                       | 2019 DOI<br>10.59350/v7xr3-6sn63<br>(open)       |
| [@baggerly2024importanc<br>e] | The Importance of<br>Reproducible Research in<br>High-Throughput Biology                  | 2024 DOI 10.52519/00193<br>(open)                |
| [@anon2019introduction]       | Introduction                                                                              | 2019 DOI<br>10.2307/j.ctvpb3xkg.6<br>(open)      |
| [@anon2019tables]             | Tables                                                                                    | 2019 DOI<br>10.1525/9780520969230-<br>002 (open) |
| [@poldrack2019case]           | Case Study 29:<br>Developing a<br>Reproducible Workflow<br>for Large-Scale<br>Phenotyping | 2019 DOI<br>10.1525/9780520967779-<br>038 (open) |
| [@anon2019what]               | What Is Ethical Research?                                                                 | 2019 DOI<br>10.2307/j.ctvpb3xkg.7<br>(open)      |

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## reproducible research XI

*Per-paper synthesis omitted — no LLM Contribution paragraphs are present in the deep-search outputs (set `project_config.deep_search.llm_per_paper: true` and ensure Ollama is reachable to populate this section).*

*Composition summary: 3 keywords · 300 unique papers · 300 per-paper notes integrated · 300 BibTeX entries · 0 key(s) missing from bib.*